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PHASE 01 — SYSTEM OVERVIEW

1. Abstract

The Art Gallery Management System (AGMS) is a database-driven digital platform designed to streamline the management of artists, artworks, customers, exhibitions, transactions, and gallery operations. The system eliminates traditional manual procedures and introduces a structured, automated environment for managing artwork catalogs, artist registrations, customer purchases, exhibition scheduling, and administrative functions.

This project applies essential database concepts including Entity–Relationship (ER) modeling, schema refinement, normalization, and SQL-based implementation. AGMS adopts a multi-user role-based framework—providing distinct interfaces for administrators, artists, and customers—to ensure data security, operational efficiency, and clear workflow separation.

The report includes system overview, database design, relational schema, SQL queries, normalization steps, progress documentation, and evaluation. AGMS demonstrates how database technologies can improve usability, consistency, accuracy, and scalability in digital gallery environments.

2. Introduction

Art galleries preserve, promote, and exhibit artistic works created by painters, sculptors, and digital artists. Traditionally, these operations are maintained manually using paper-based logs or spreadsheets, leading to inefficiencies and data inaccuracies. Tasks such as storing artwork information, managing artist portfolios, recording sales transactions, and organizing exhibitions become increasingly complex as gallery size grows.

The Art Gallery Management System (AGMS) is developed to address these challenges through a comprehensive and automated solution. Built on a relational database architecture, AGMS manages and organizes artwork data, artist information, customer interactions, and exhibition workflows efficiently. The system enhances accessibility, simplifies operations, reduces redundancy, and ensures data consistency.

3. Purpose of the System

The purpose of AGMS is to streamline and modernize gallery operations through digital automation. The system aims to:

- Provide a centralized database for storing artwork, artist, customer, and exhibition details
- Reduce manual workload and minimize errors

- Enable online browsing and purchasing of artworks
- Facilitate artist registration and portfolio maintenance
- Support administrative control over gallery activities
- Ensure secure and structured data handling

The system ultimately makes gallery management more efficient, transparent, and user-friendly.

4. System Objectives

Functional Objectives

- Provide a seamless platform for customers to browse and purchase artworks
- Enable artists to upload, update, and manage their artwork portfolios
- Allow administrators to monitor system-wide activities
- Support exhibition creation and the participation of artists
- Maintain purchase records, transaction logs, and customer history

Technical Objectives

- Implement a normalized relational database to minimize redundancy
- Use ER modeling for conceptual design
- Apply integrity constraints to ensure data correctness
- Optimize SQL queries for efficient data retrieval
- Maintain role-based secure access

5. Project Scope

The AGMS covers a wide range of gallery-related functionalities:

Admin

- Manage users (artists, customers)
- Moderate artworks
- Set up exhibitions
- View and analyze reports

Artist

- Register as an artist
- Upload artworks
- Manage portfolios
- Track sales and exhibition participation

Customer

- Browse artworks
- Purchase artworks
- Maintain purchase history

Database Scope

- Artwork details (image, title, description, category, price)
- Artist data (biography, contact, portfolio)
- Customer accounts and orders
- Transaction records
- Exhibition and event scheduling

Current System Limitations

- Payment gateway not integrated
- No automated recommendation engine
- Media storage limited to local directory

6. User Types & Roles

The system distinguishes three primary user roles:

1. Admin

Key Responsibilities:

- Approve artist registrations
- Moderate artwork submissions

- Manage exhibitions
- Generate system reports
- Oversee user activity and logs

Privileges:

Full access to the entire system; can modify or delete any record.

2. Artist

Key Responsibilities:

- Register and maintain profile
- Upload and update artworks
- Track purchases made by customers
- Join exhibitions

Privileges:

Can manage only their own artworks and profile.

3. Customer (Buyer)

Key Responsibilities:

- Browse, view, and purchase artworks
- Add items to cart
- Track order history

Privileges:

Limited to browsing and purchasing functionalities.

7. Functional Requirements (High-Level)

Admin Requirements

- Admin authentication
- Artist management
- Artwork moderation

- Exhibition scheduling
- Report generation

Artist Requirements

- Registration & login
- Artwork upload/edit/delete
- Profile management
- Exhibition participation

Customer Requirements

- Browse artwork with filtering
- View artwork details
- Add to cart & checkout
- Maintain order history

8. System Features Overview

Key features implemented in AGMS include:

- Image-based artwork catalog
- Artist portfolio module
- Cart and checkout system
- Order management
- Exhibition module with admin control
- Reporting dashboard
- Secure login and role-based access

9. System Workflow Description

Artist Workflow

Register → Upload Artwork → Artwork Published → Exhibition Participation

Customer Workflow

Browse Artwork → View Details → Add to Cart → Checkout → Order Confirmation

Admin Workflow

Login → Manage Users & Artworks → Create Exhibitions → View Reports → Maintain System

10. Use Case Diagram (Text Explanation)

Actors

- Admin
- Artist
- User

Core Use Cases

- Manage Artist (Admin)
- Manage Artwork (Admin/Artist)
- Manage Exhibition (Admin)
- Browse Artwork (Customer)
- Purchase Artwork (Customer)
- Track Orders (Customer)

Include / Extend Relationships

- “Verify Payment” is included in “Purchase Artwork”
- “View Artwork Details” extends “Browse Artwork”

11. Screens / Page Overview (with Figure Placeholders)

Figure 1: Home Page

The Home Page serves as the main entry point to the Art Gallery Management System, showcasing featured artworks and trending pieces. Users can browse visually rich collections, search for artists or paintings, and navigate to login or sign-up options. It provides an inviting gallery-like experience, encouraging exploration and engagement with the platform.

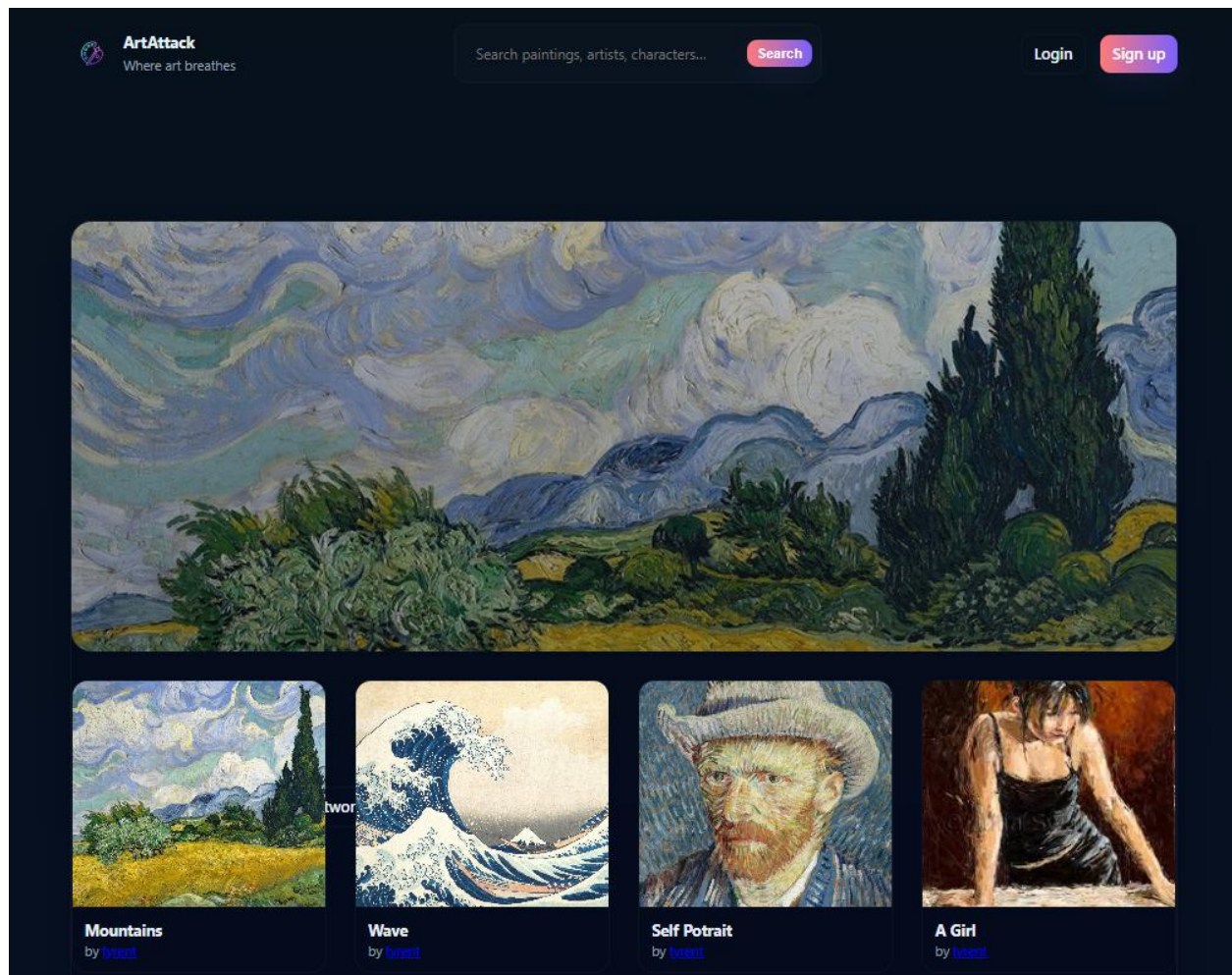
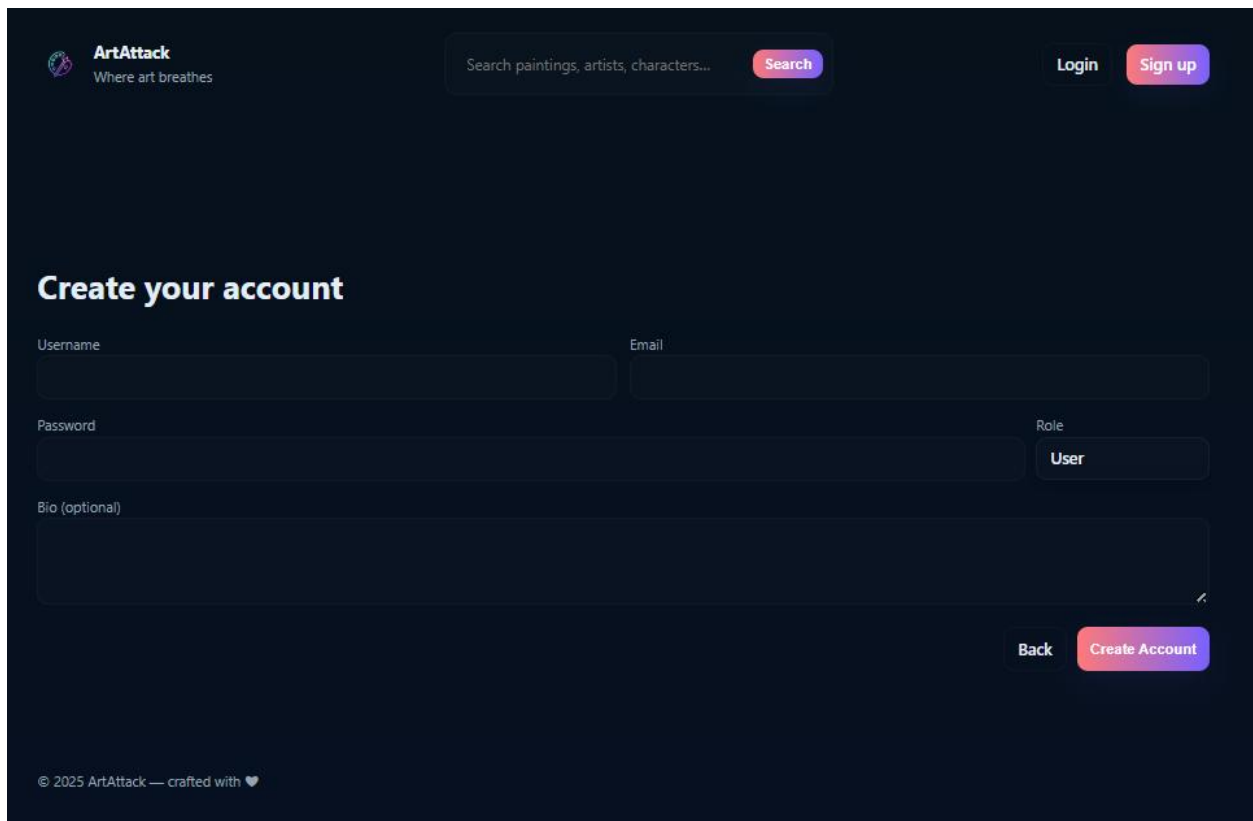


Figure 2: Registration Page

The Sign Up Page allows new users to create an account by providing a username, email, password, and optional bio. It enables role selection, determining whether the user registers as an artist or customer. This page ensures secure onboarding, granting personalized access to platform features such as artwork uploads, browsing, and purchases.



The screenshot displays the 'Create your account' registration page for ArtAttack. The page has a dark blue background. At the top left is the ArtAttack logo with the tagline 'Where art breathes'. In the top center is a search bar with the placeholder text 'Search paintings, artists, characters...' and a 'Search' button. At the top right are 'Login' and 'Sign up' buttons. The main heading 'Create your account' is in white. Below it are input fields for 'Username', 'Email', 'Password', and 'Bio (optional)'. To the right of the 'Password' field is a 'Role' dropdown menu currently set to 'User'. At the bottom right are 'Back' and 'Create Account' buttons. The footer at the bottom left reads '© 2025 ArtAttack — crafted with ♥'.

ArtAttack
Where art breathes

Search paintings, artists, characters... Search

Login Sign up

Create your account

Username Email

Password Role
User

Bio (optional)

Back Create Account

© 2025 ArtAttack — crafted with ♥

Figure 3: Login Page

The Login Page provides secure access for returning users through email or username and password authentication. It supports a “Remember me” option and links for account creation or password recovery. This page ensures only authorized customers, artists, and admins can access personalized dashboards, upload artworks, manage profiles, or perform system operations.

ArtAttack
Where art breathes

Search paintings, artists, characters... **Search**

Login **Sign up**

Welcome back

Log in to continue to ArtAttack.

Email or username

Password

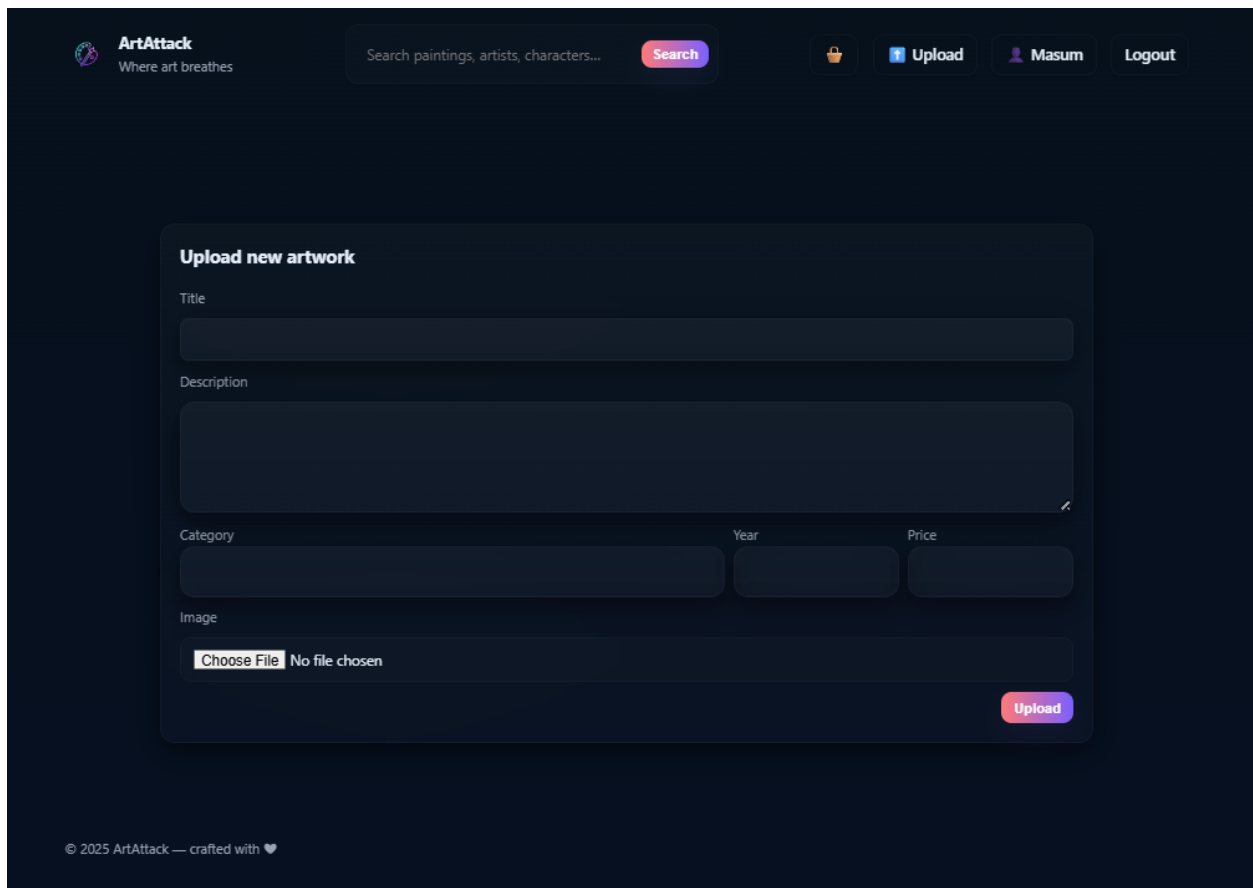
☐ Remember me [Forgot?](#)

Log in **Create account**

© 2025 ArtAttack — crafted with ❤️

Figure 4: Artwork Upload Page

The Artwork Upload Page enables artists to submit new artworks by providing essential details such as title, description, category, year, price, and image file. It simplifies portfolio management, allowing artists to expand their collections on the platform. The page ensures structured data entry and smooth integration of new artworks into the gallery system.



The screenshot displays the 'ArtAttack' website's artwork upload interface. The header features the 'ArtAttack' logo with the tagline 'Where art breathes', a search bar with the placeholder 'Search paintings, artists, characters...' and a 'Search' button, a shopping cart icon, and user links for 'Upload', 'Masum', and 'Logout'. The main content area is titled 'Upload new artwork' and contains a form with the following fields: 'Title' (a single-line text input), 'Description' (a multi-line text area), 'Category' (a dropdown menu), 'Year' (a single-line text input), 'Price' (a single-line text input), and 'Image' (a file upload button labeled 'Choose File' with the text 'No file chosen' next to it). An 'Upload' button is positioned at the bottom right of the form. The footer includes the copyright notice '© 2025 ArtAttack — crafted with ♥'.

Figure 5: Artwork Browsing Page

The Artwork Browsing Page allows users to explore all available artworks through a visually appealing grid layout. It includes search and category filters to refine results based on title, description, or genre. Users can view artist names, preview artwork details, and seamlessly navigate to individual artwork pages for further information or purchase options.

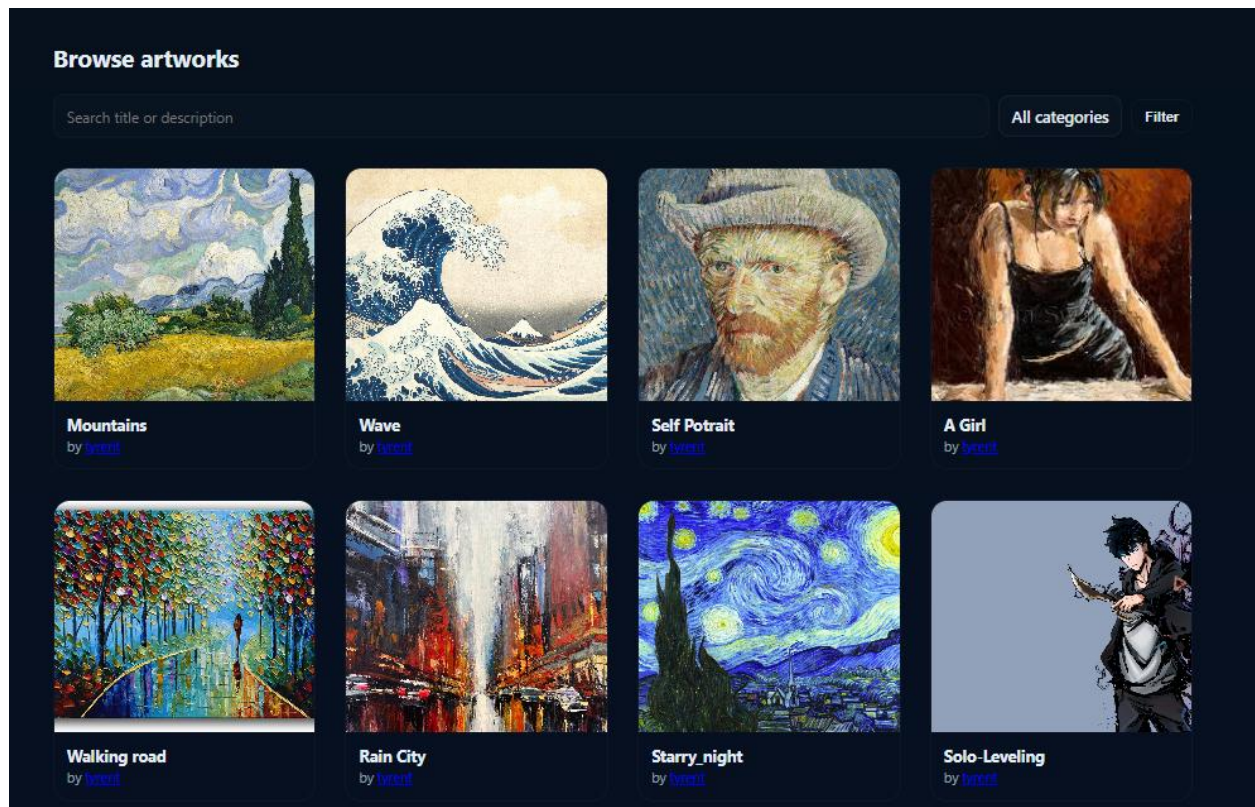


Figure 6: Artwork Details View

The Artwork Details Page provides an in-depth view of a selected artwork, including its image, title, artist, price, and description. Users can like the artwork, add it to their cart, or leave ratings and reviews. The page centralizes interactions, offering a personalized experience while encouraging engagement and potential purchases.

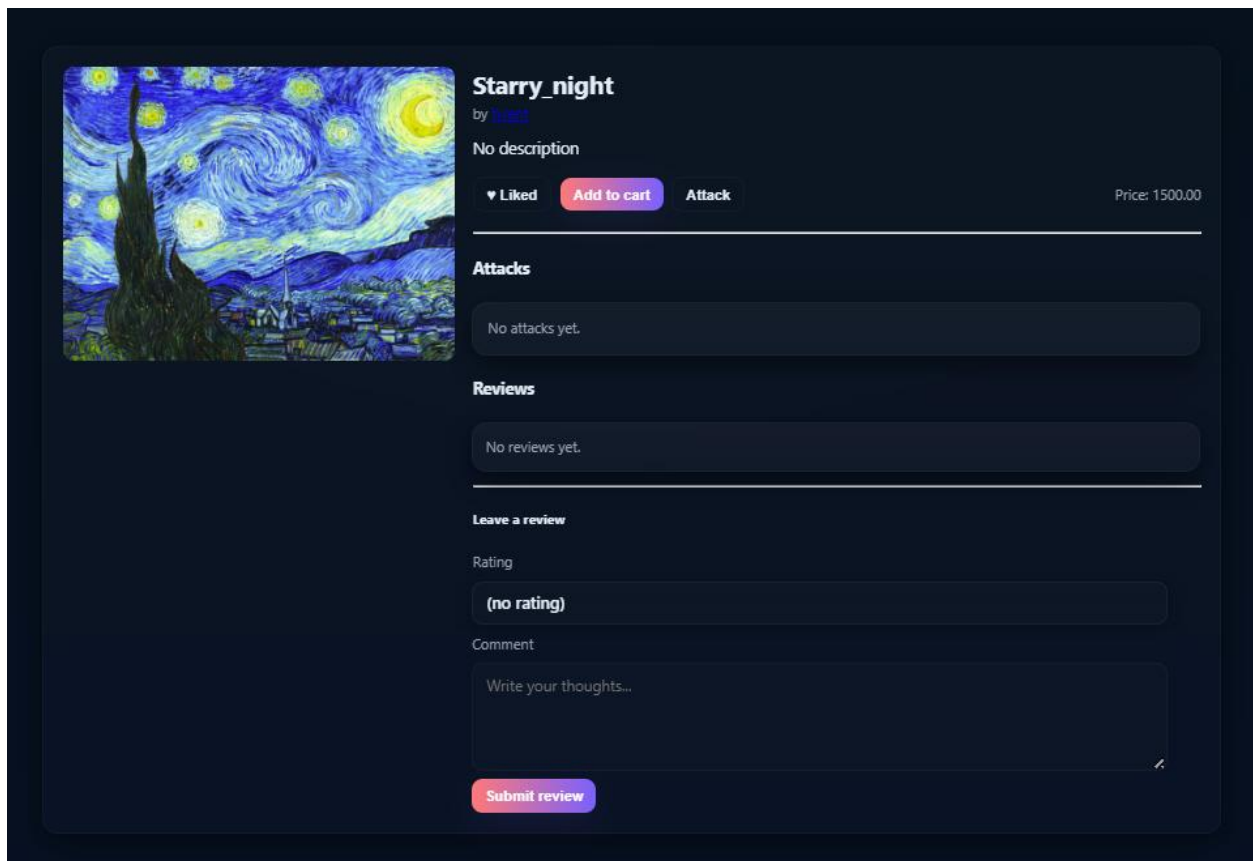


Figure 7: Cart Page

The Cart Page displays artworks selected for purchase, along with their prices, artist names, and thumbnails. Users can remove items, view the subtotal, and enter a delivery address before proceeding to checkout. This page streamlines the purchasing process by summarizing all selected artworks in one place and enabling a smooth transaction flow.

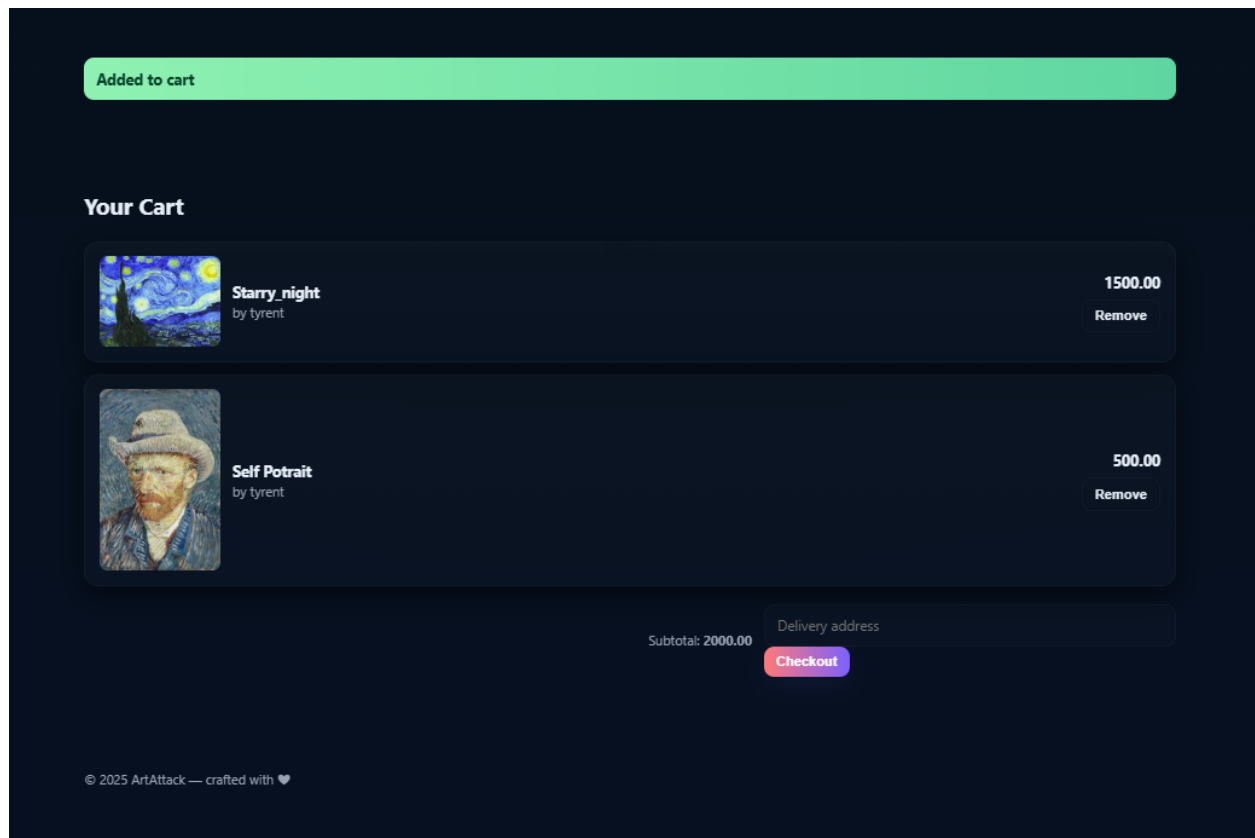
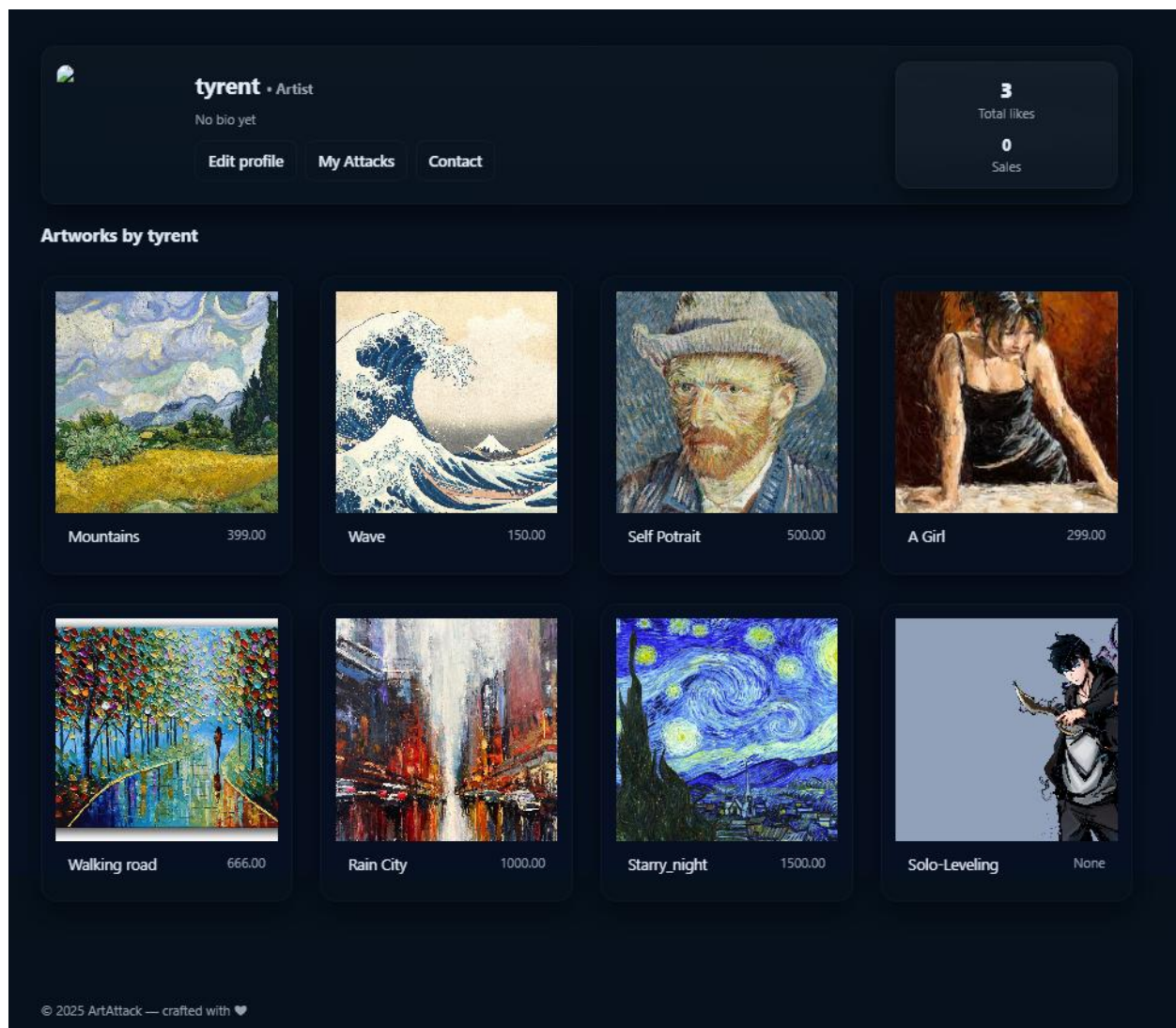


Figure 8: Artists Profile Page

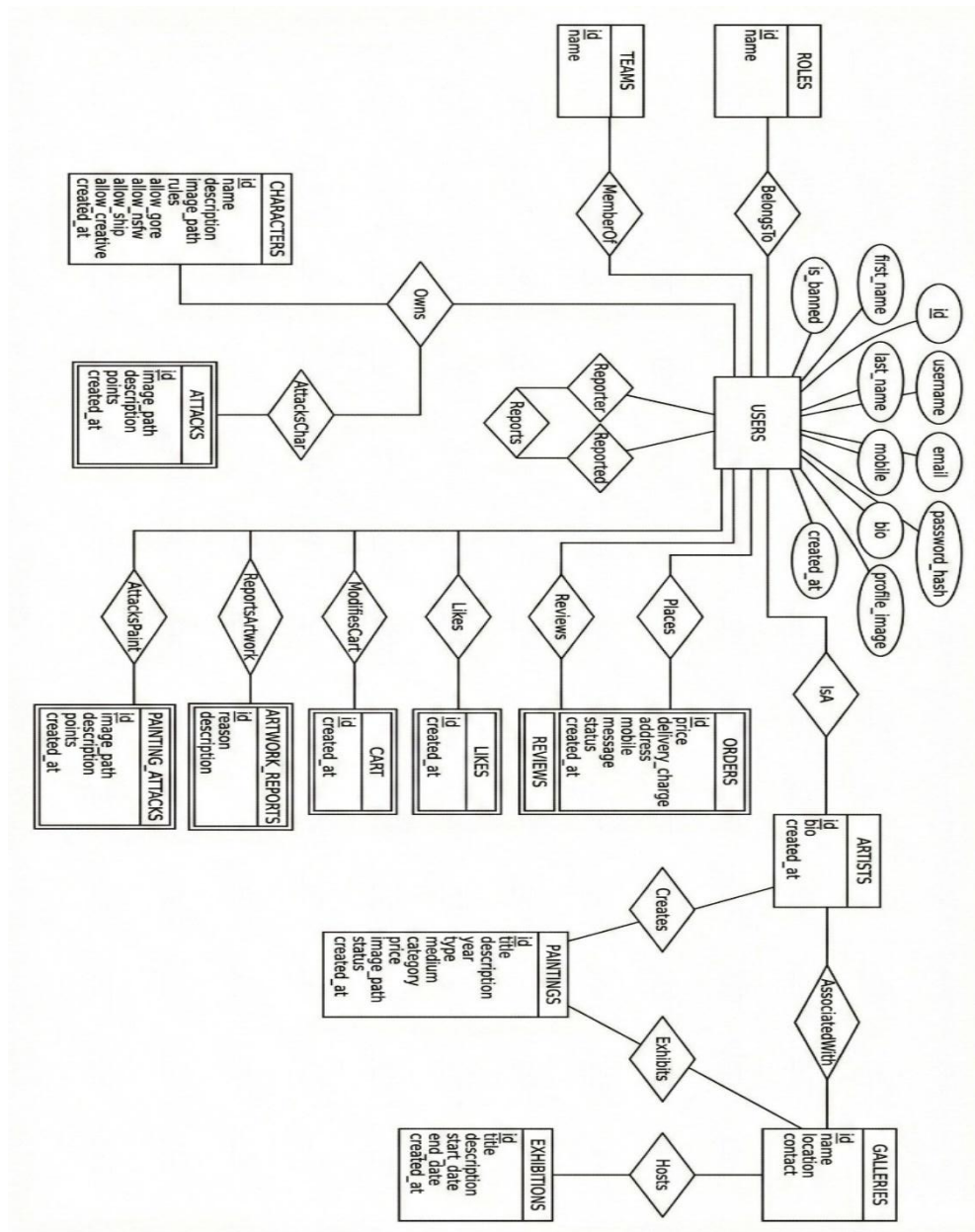
The Artist Profile Page showcases an artist's personal information, total likes, and sales performance. It displays all artworks uploaded by the artist in a clean gallery layout, each with pricing details. Users can view the artist's portfolio, while artists can edit their profile, manage uploaded artworks, and interact with viewers through available contact options.



PHASE 02 — DATABASE DESIGN

2.1 Entity–Relationship (ER) Diagram — Conceptual Explanation

The Art Gallery Management System (AGMS) stores and manages a variety of interdependent data related to artists, artworks, customers, transactions, and exhibitions. The ER diagram represents the conceptual model of how these entities relate to one another within the system.



The system contains the following core entities:

1. **Artist**
2. **Artwork**
3. **Customer**
4. **Orders**
5. **Order_Items** (if multiple artworks per order)
6. **Exhibition**
7. **Exhibition_Artwork** (bridge entity)
8. **Admin**
9. **Category**

2.1.1 List of Entities & Their Descriptions

1. Artist

Represents registered artists who upload artworks.

Attribute	Description
Artist_ID (PK)	Unique identifier
Name	Full name
Email	Unique contact email
Password	Encrypted password
Biography	Artist's description
Profile_Image	Image path
Join_Date	Registration date

2. Artwork

Represents artworks uploaded by artists.

Attribute	Description
Artwork_ID (PK)	Unique artwork ID
Artist_ID (FK)	References Artist
Title	Artwork title
Description	Detailed info
Category_ID (FK)	Artwork category
Year	Creation year
Price	Artwork price
Image_Path	Stored file path
Status	Available/Sold
Upload_Date	Creation timestamp

3. Customer

Attribute	Description
Customer_ID (PK)	Unique buyer ID
Name	Buyer full name
Email	Unique login email
Password	Encrypted password
Address	Delivery address
Phone	Contact number
Join_Date	Registration date

4. Orders

Attribute	Description
Order_ID (PK)	Unique order number
Customer_ID (FK)	Purchase initiator
Order_Date	Date/time of order
Total_Amount	Price total
Status	Pending/Completed/Cancelled

5. Order_Items

Supports multiple artworks per order.

Attribute	Description
Order_Item_ID (PK)	Unique entry
Order_ID (FK)	References Orders
Artwork_ID (FK)	Artwork purchased
Price	Snapshot price
Quantity	Default 1

6. Category

Represents artwork categories (Portrait, Abstract, Landscape, Modern, etc.)

Attribute	Description
Category_ID (PK)	Unique ID
Category_Name	Category name

7. Exhibition

Attribute	Description
Exhibition_ID (PK)	Unique event ID
Title	Exhibition name
Description	Event description
Start_Date	From date
End_Date	To date

8. Exhibition_Artwork (Bridge Entity)

Supports many-to-many relationship between Exhibition and Artwork.

Attribute	Description
Exhibition_ID (FK)	References Exhibition
Artwork_ID (FK)	References Artwork

Composite PK: (Exhibition_ID, Artwork_ID)

9. Admin

Attribute	Description
Admin_ID (PK)	Unique admin ID
Name	Admin full name
Email	Login credential
Password	Encrypted password

2.1.2 Relationships Description

1. Artist → Artwork

Relationship: 1-to-Many

- One artist can upload multiple artworks.
- Each artwork belongs to exactly one artist.

2. Customer → Orders

Relationship: 1-to-Many

- A customer can place multiple orders.
- An order belongs to a single customer.

3. Orders → Order_Items → Artwork

Relationship:

- Orders to Order_Items → 1-to-Many
- Artwork to Order_Items → 1-to-Many

4. Exhibition ↔ Artwork

Relationship: Many-to-Many via Exhibition_Artwork

- An artwork can be displayed in multiple exhibitions.
- An exhibition can include multiple artworks.

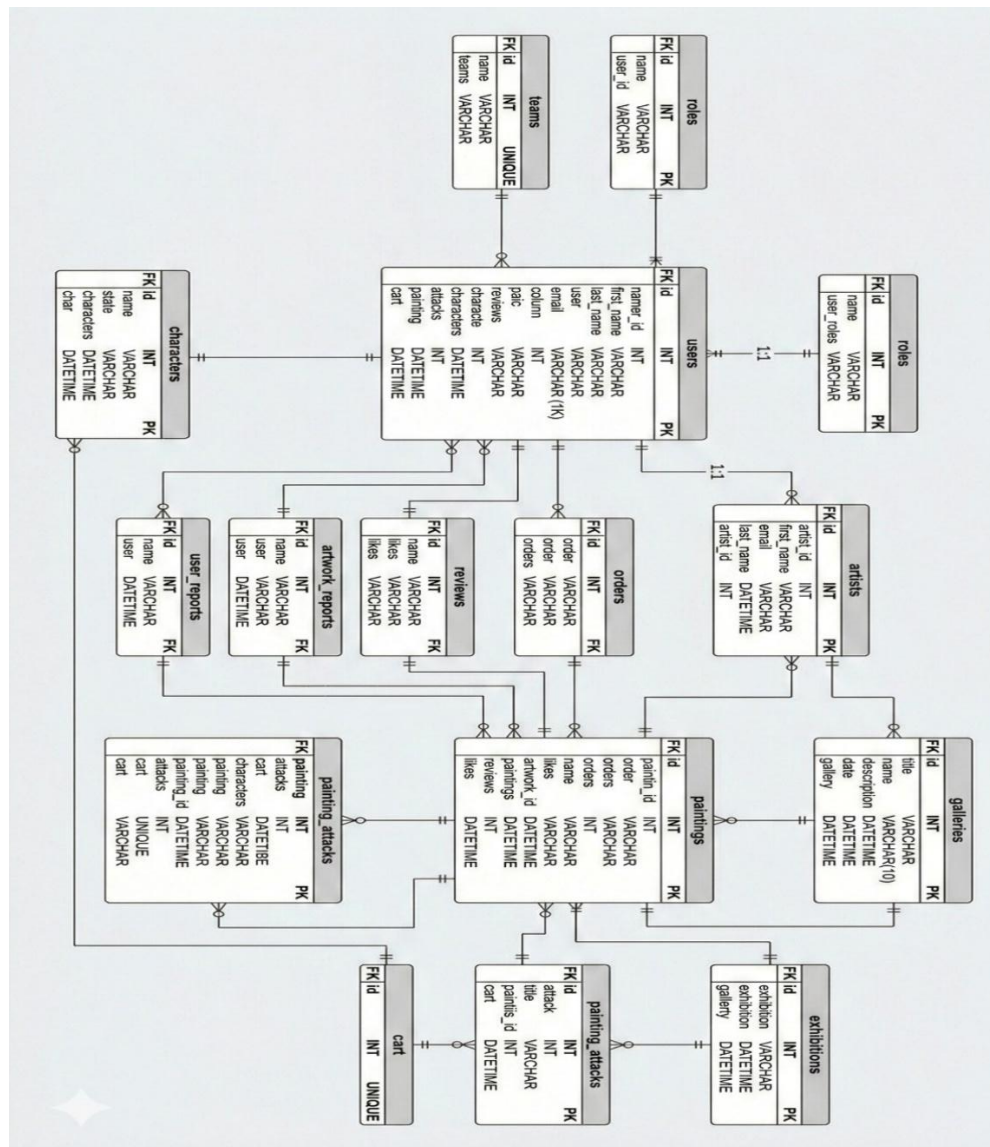
5. Admin → System Entities

Admin has supervisory (not relational) authority.

Admin can:

- Approve artists
- Remove artworks
- Manage exhibitions

2.2 Full Relational Schema



Admin(
 Admin_ID PK,
 Name,
 Email UNIQUE,
 Password
)

Artist(

Artist_ID PK,
Name,
Email UNIQUE,
Password,
Biography,
Profile_Image,
Join_Date
)

Artwork(

Artwork_ID PK,
Artist_ID FK → Artist.Artist_ID,
Category_ID FK → Category.Category_ID,
Title,
Description,
Year,
Price,
Image_Path,
Status,
Upload_Date
)

Customer(

Customer_ID PK,
Name,
Email UNIQUE,
Password,
Address,
Phone,
Join_Date
)

Orders(

Order_ID PK,
Customer_ID FK → Customer.Customer_ID,
Order_Date,
Total_Amo)

Order_Items(

Order_Item_ID PK,
Order_ID FK → Orders.Order_ID,
Artwork_ID FK → Artwork.Artwork_ID,
Price,
Quantity
)

Category(

Category_ID PK,
Category_Name
)

Exhibition(

Exhibition_ID PK,
Title,
Description,
Start_Date,
End_Date
)

Exhibition_Artwork(

Exhibition_ID FK → Exhibition.Exhibition_ID,
Artwork_ID FK → Artwork.Artwork_ID,
PRIMARY KEY (Exhibition_ID, Artwork_ID)
)

2.3 Integrity Constraints

Primary Key Constraints

- Every entity has a unique primary key.

Foreign Key Constraints

Examples:

Artwork.Artist_ID → Artist.Artist_ID

Artwork.Category_ID → Category.Category_ID

Orders.Customer_ID → Customer.Customer_ID

Order_Items.Order_ID → Orders.Order_ID

Order_Items.Artwork_ID → Artwork.Artwork_ID

Exhibition_Artwork.Exhibition_ID → Exhibition.Exhibition_ID

Exhibition_Artwork.Artwork_ID → Artwork.Artwork_ID

Domain Constraints

- Price > 0
- Quantity >= 1
- Status IN ('Available', 'Sold', 'Pending')
- Email format validation
- Year BETWEEN 1500 AND current_year

Entity Integrity

- No NULL allowed in primary keys
- Non-null required for essential attributes (Name, Email, Title, Price)

Referential Integrity

- Cannot delete Artist if Artwork exists
- Cannot delete Customer if Orders exist
- Cascade deletion optional for:

ON DELETE CASCADE (Order_Items)

2.4 Normalization

1st Normal Form (1NF)

The given relation is in 1st Normal Form because it meets the following criteria:

1. All cells contain atomic (indivisible) values.
2. Entries in each column are of the same data type.
3. Each record is unique; there are no duplicate rows.
4. Attribute names are unique and meaningful.

2nd Normal Form (2NF)

The given relation is in 2nd Normal Form because it satisfies these conditions:

1. The table is already in 1st Normal Form.
2. There is no partial dependency; no subset of a candidate key determines a non-key attribute.
3. All non-key attributes are fully dependent on the entire primary key.

3rd Normal Form (3NF)

The given relation is in 3rd Normal Form because:

1. The table is already in 2nd Normal Form.
2. There is no transitive dependency; a non-prime attribute does not depend on another non-prime attribute.
3. For every functional dependency, the left-hand side is a key, and the right-hand side is a non-prime attribute.

Thus, all columns are determined only by the primary key and no other non-key attributes.

1. Admin table:

```
mysql> DESC Admin;
```

Field	Type	Null	Key	Default	Extra
Admin_ID	int	NO	PRI	NULL	auto_increment
Name	varchar(100)	NO		NULL	
Email	varchar(100)	NO	UNI	NULL	
Password	varchar(255)	NO		NULL	

4 rows in set (0.00 sec)

Normalization Check Table for Admin

Normal Form	✓ / ✗	Explanation
First Normal Form (1NF)	✓	All fields are atomic (no repeating groups). Each column holds a single value (e.g., one Email per row). Each row is uniquely identified by the primary key Admin_ID.
Second Normal Form (2NF)	✓	Table is in 1NF and the primary key is a single attribute (Admin_ID). There is no partial dependency because there is no composite key — every non-key attribute (Name, Email, Password) depends on the whole (single) primary key.
Third Normal Form (3NF)	✓	Table is in 2NF and has no transitive dependencies. None of the non-key attributes (Name, Email, Password) functionally determines another non-key attribute. They all depend only on Admin_ID. Therefore, 3NF holds.

Conclusion:

The Admin table is in Third Normal Form (3NF).

2. Artist table:

```
mysql> DESC Artist;
```

Field	Type	Null	Key	Default	Extra
Artist_ID	int	NO	PRI	NULL	auto_increment
Name	varchar(100)	NO		NULL	
Email	varchar(100)	NO	UNI	NULL	
Password	varchar(255)	NO		NULL	
Biography	text	YES		NULL	
Profile_Image	varchar(255)	YES		NULL	
Join_Date	date	YES		NULL	

7 rows in set (0.00 sec)

Normalization Check Table for Artist

Normal Form	✓ / ✗	Explanation
First Normal Form (1NF)	✓	All fields are atomic (no repeating groups). Each column holds a single value (one email, one name, one join date per row). Each row is uniquely identified by the primary key Artist_ID.
Second Normal Form (2NF)	✓	The table is in 1NF and the primary key is a single attribute (Artist_ID). There is no partial dependency because there is no composite key; all non-key attributes (Name, Email, Password, Biography, Profile_Image, Join_Date) depend on the whole (single) primary key.
Third Normal Form (3NF)	✓	The table is in 2NF and there are no transitive dependencies. None of the non-key attributes (e.g., Email, Biography, Join_Date) functionally determines another non-key attribute. All depend directly on Artist_ID. Therefore, 3NF holds.

Conclusion:

The Artist table is in Third Normal Form (3NF).

3. Customer table:

```
mysql> DESC Customer;
```

Field	Type	Null	Key	Default	Extra
Customer_ID	int	NO	PRI	NULL	auto_increment
Name	varchar(100)	NO		NULL	
Email	varchar(100)	NO	UNI	NULL	
Password	varchar(255)	NO		NULL	
Address	varchar(255)	YES		NULL	
Phone	varchar(30)	YES		NULL	
Join_Date	date	YES		NULL	

7 rows in set (0.00 sec)

Normalization Check Table for Customer

Normal Form	✓ / ✗	Explanation
First Normal Form (1NF)	✓	All attributes are atomic (no repeating groups). Each field holds a single value, such as one email, one address, one phone per row. Each row is uniquely identified by Customer_ID.
Second Normal Form (2NF)	✓	The table is in 1NF and the primary key is a single attribute (Customer_ID). There is no partial dependency because there is no composite key — all non-key attributes (Name, Email, Password, Address, Phone, Join_Date) depend on the whole (single) primary key.
Third Normal Form (3NF)	✓	The table is in 2NF and has no transitive dependencies. No non-key attribute (e.g., Email, Address, Phone) functionally determines another non-key attribute. All depend directly on Customer_ID. Therefore, 3NF holds.

Conclusion:

The Customer table is in Third Normal Form (3NF).

4. Category table:

```
mysql> DESC Category;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| Category_ID    | int           | NO   | PRI | NULL    | auto_increment |
| Category_Name  | varchar(100)  | NO   |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

Normalization Check Table for Category

Normal Form	✓ / ✗	Explanation
First Normal Form (1NF)	✓	All fields are atomic and there are no repeating groups. Each row has a single Category_ID and a single Category_Name. The primary key Category_ID uniquely identifies each category.
Second Normal Form (2NF)	✓	The table is in 1NF and the primary key is a single attribute (Category_ID). There is no partial dependency because there is no composite key; the only non-key attribute Category_Name depends on the whole (single) primary key.
Third Normal Form (3NF)	✓	The table is in 2NF and has no transitive dependencies. Category_Name does not determine any other non-key attribute (there are none). It depends only on Category_ID. Therefore, 3NF holds.

Conclusion:

The Category table is in Third Normal Form (3NF).

5. Artwork table:

```
mysql> DESC Artwork;
```

Field	Type	Null	Key	Default	Extra
Artwork_ID	int	NO	PRI	NULL	auto_increment
Artist_ID	int	NO	MUL	NULL	
Category_ID	int	YES	MUL	NULL	
Title	varchar(150)	NO		NULL	
Description	text	YES		NULL	
Year	int	YES		NULL	
Price	decimal(10,2)	NO		NULL	
Image_Path	varchar(255)	YES		NULL	
Status	varchar(20)	YES		Available	
Upload_Date	date	YES		NULL	

10 rows in set (0.00 sec)

Normalization Check Table for Artwork

Normal Form ✓ / ✗ Explanation

First Normal Form (1NF) ✓ All fields are atomic and there are no repeating groups. Each row represents one artwork with a single Artist_ID, Category_ID, Title, Price, etc. The primary key Artwork_ID uniquely identifies every row.

Second Normal Form (2NF) ✓ The table is in 1NF and the primary key is a single attribute (Artwork_ID). There is no partial dependency because there is no composite key — every non-key attribute (Artist_ID, Category_ID, Title, Description, Year, Price, Image_Path, Status, Upload_Date) depends on the whole (single) primary key.

Third Normal Form (3NF) ✓ The table is in 2NF and has no transitive dependencies. Non-key attributes do not determine other non-key attributes: for example, Category_ID or Artist_ID do not store Category_Name or Artist_Name here (those are in separate tables). All non-key attributes depend only on Artwork_ID. Therefore, 3NF holds.

Conclusion:

The Artwork table is in Third Normal Form (3NF).

6. Orders table:

```
mysql> DESC Orders;
```

Field	Type	Null	Key	Default	Extra
Order_ID	int	NO	PRI	NULL	auto_increment
Customer_ID	int	NO	MUL	NULL	
Order_Date	datetime	YES		NULL	
Total_Amount	decimal(10,2)	NO		0.00	
Status	varchar(20)	YES		Pending	

5 rows in set (0.00 sec)

Normalization Check Table for Orders

Normal Form	✓ / ✗	Explanation
First Normal Form (1NF)	✓	All fields are atomic; there are no repeating groups. Each row stores a single order with one Customer_ID, one Order_Date, one Total_Amount, and one Status. The primary key Order_ID uniquely identifies each order.
Second Normal Form (2NF)	✓	The table is already in 1NF and the primary key is a single attribute (Order_ID). There is no partial dependency because there is no composite key. Every non-key attribute (Customer_ID, Order_Date, Total_Amount, Status) depends on the whole (single) primary key.
Third Normal Form (3NF)	✓	The table is in 2NF and has no transitive dependencies. Non-key attributes do not determine each other: for example, Customer_ID does not store customer details here (they are in the Customer table). All non-key attributes depend only on Order_ID. Therefore, 3NF holds.

Conclusion:

The Orders table is in Third Normal Form (3NF).

7. Order_Items table:

```
mysql> DESC Order_Items;
```

Field	Type	Null	Key	Default	Extra
Order_Item_ID	int	NO	PRI	NULL	auto_increment
Order_ID	int	NO	MUL	NULL	
Artwork_ID	int	NO	MUL	NULL	
Price	decimal(10,2)	NO		NULL	
Quantity	int	NO		1	

5 rows in set (0.00 sec)

Normalization Check Table for Order_Items

Normal Form	✓ / ✗	Explanation
First Normal Form (1NF)	✓	All fields are atomic and there are no repeating groups. Each row represents a single line item: one Order_ID, one Artwork_ID, one Price, and one Quantity. The primary key Order_Item_ID uniquely identifies each row.
Second Normal Form (2NF)	✓	The table is in 1NF and has a single-attribute primary key (Order_Item_ID). There is no partial dependency, since there is no composite primary key — all non-key attributes (Order_ID, Artwork_ID, Price, Quantity) depend on the whole (single) primary key.
Third Normal Form (3NF)	✓	The table is in 2NF and has no transitive dependencies. Non-key attributes do not determine other non-key attributes: for example, Order_ID does not store order details here, and Artwork_ID does not store artwork details here. These are stored in Orders and Artwork tables respectively. Therefore, all non-key attributes depend only on Order_Item_ID, so 3NF holds.

Conclusion:

The Order_Items table is in Third Normal Form (3NF).

8. Exhibition table:

```
mysql> DESC Exhibition;
```

Field	Type	Null	Key	Default	Extra
Exhibition_ID	int	NO	PRI	NULL	auto_increment
Title	varchar(150)	NO		NULL	
Description	text	YES		NULL	
Start_Date	date	NO		NULL	
End_Date	date	NO		NULL	

5 rows in set (0.00 sec)

Normalization Check Table for Exhibition

Normal Form	✓ / ✗	Explanation
First Normal Form (1NF)	✓	All fields are atomic and there are no repeating groups. Each row stores a single exhibition with one Title, one Start_Date, and one End_Date. The primary key Exhibition_ID uniquely identifies each exhibition.
Second Normal Form (2NF)	✓	The table is in 1NF and the primary key is a single attribute (Exhibition_ID). There is no partial dependency, because there is no composite key — all non-key attributes (Title, Description, Start_Date, End_Date) depend on the entire (single) primary key.
Third Normal Form (3NF)	✓	The table is in 2NF and has no transitive dependencies. Non-key attributes do not determine other non-key attributes; e.g., Title or Start_Date does not functionally determine End_Date within this table. All non-key attributes depend only on Exhibition_ID. Therefore, 3NF holds.

Conclusion:

The Exhibition table is in Third Normal Form (3NF).

9. Exhibition_Artwork table:

```
mysql> DESC Exhibition_Artwork;
+-----+-----+-----+-----+-----+
| Field          | Type  | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+
| Exhibition_ID  | int   | NO   | PRI | NULL    |       |
| Artwork_ID     | int   | NO   | PRI | NULL    |       |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

(Primary key is composite: (Exhibition_ID, Artwork_ID), both normally also foreign keys to Exhibition and Artwork.)

Normalization Check Table for Exhibition_Artwork

Normal Form	✓ / ✗	Explanation
First Normal Form (1NF)	✓	All fields are atomic, with no repeating groups. Each row links one exhibition to one artwork using Exhibition_ID and Artwork_ID. The composite primary key (Exhibition_ID, Artwork_ID) uniquely identifies each row.
Second Normal Form (2NF)	✓	The table is in 1NF and the primary key is composite (Exhibition_ID, Artwork_ID). There are no non-key attributes at all, so it is impossible to have a partial dependency of a non-key attribute on part of the key. Therefore, 2NF holds trivially.
Third Normal Form (3NF)	✓	The table is in 2NF and has no non-key attributes, so transitive dependencies cannot exist. Every attribute is part of the key itself. Hence, the table satisfies 3NF by definition.

Conclusion:

The Exhibition_Artwork table is in Third Normal Form (3NF).

2.5 SQL Statements

Create Database

```
CREATE DATABASE AGMS;
```

```
USE AGMS;
```

Admin Table

```
CREATE TABLE Admin (  
    Admin_ID INT AUTO_INCREMENT PRIMARY KEY,  
    Name VARCHAR(100),  
    Email VARCHAR(100) UNIQUE,  
    Password VARCHAR(255)  
);
```

Artist Table

```
CREATE TABLE Artist (  
    Artist_ID INT AUTO_INCREMENT PRIMARY KEY,  
    Name VARCHAR(100),  
    Email VARCHAR(100) UNIQUE,  
    Password VARCHAR(255),  
    Biography TEXT,  
    Profile_Image VARCHAR(255),  
    Join_Date DATE  
);
```

Category Table

```
CREATE TABLE Category (  
    Category_ID INT AUTO_INCREMENT PRIMARY KEY,  
    Category_Name VARCHAR(100)  
);
```

Artwork Table

```
CREATE TABLE Artwork (  
    Artwork_ID INT AUTO_INCREMENT PRIMARY KEY,  
    Artist_ID INT,  
    Category_ID INT,  
    Title VARCHAR(150),  
    Description TEXT,  
    Year INT,  
    Price DECIMAL(10,2),  
    Image_Path VARCHAR(255),  
    Status VARCHAR(20),  
    Upload_Date DATE,  
    FOREIGN KEY (Artist_ID) REFERENCES Artist(Artist_ID),  
    FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID)  
);
```

Customer Table

```
CREATE TABLE Customer (  
    Customer_ID INT AUTO_INCREMENT PRIMARY KEY,  
    Name VARCHAR(100),  
    Email VARCHAR(100) UNIQUE,
```

```
Password VARCHAR(255),  
Address VARCHAR(255),  
Phone VARCHAR(30),  
Join_Date DATE  
);
```

Orders Table

```
CREATE TABLE Orders (  
    Order_ID INT AUTO_INCREMENT PRIMARY KEY,  
    Customer_ID INT,  
    Order_Date DATE,  
    Total_Amount DECIMAL(10,2),  
    Status VARCHAR(20),  
    FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID)  
);
```

Order_Items Table

```
CREATE TABLE Order_Items (  
    Order_Item_ID INT AUTO_INCREMENT PRIMARY KEY,  
    Order_ID INT,  
    Artwork_ID INT,  
    Price DECIMAL(10,2),  
    Quantity INT DEFAULT 1,  
    FOREIGN KEY (Order_ID) REFERENCES Orders(Order_ID),  
    FOREIGN KEY (Artwork_ID) REFERENCES Artwork(Artwork_ID)  
);
```

Exhibition Table

```
CREATE TABLE Exhibition (  
    Exhibition_ID INT AUTO_INCREMENT PRIMARY KEY,  
    Title VARCHAR(150),  
    Description TEXT,  
    Start_Date DATE,  
    End_Date DATE  
);
```

Exhibition_Artwork Table

```
CREATE TABLE Exhibition_Artwork (  
    Exhibition_ID INT,  
    Artwork_ID INT,  
    PRIMARY KEY (Exhibition_ID, Artwork_ID),  
    FOREIGN KEY (Exhibition_ID) REFERENCES Exhibition(Exhibition_ID),  
    FOREIGN KEY (Artwork_ID) REFERENCES Artwork(Artwork_ID)  
);
```

2.6 Database tables after inserting info's

1. Admin Table

```
mysql> SELECT * FROM Admin;
```

Admin_ID	Name	Email	Password
1	System Admin	admin@agms.com	admin_hashed_password
2	Gallery Manager One	gm1@agms.com	gm1_hashed
3	Gallery Manager Two	gm2@agms.com	gm2_hashed
4	Events Coordinator	events@agms.com	events_hashed
5	Sales Supervisor	sales@agms.com	sales_hashed
6	Support Admin	support@agms.com	support_hashed
7	Content Moderator	content@agms.com	content_hashed
8	Finance Admin	finance@agms.com	finance_hashed
9	PR Admin	pr@agms.com	pr_hashed
10	Tech Admin	tech@agms.com	tech_hashed
11	Backup Admin	backup@agms.com	backup_hashed

11 rows in set (0.01 sec)

2. Artist Table

```
mysql> SELECT * FROM Artist;
```

Artist_ID	Name	Email	Password	Biography	Profile_Image	Join_Date
1	Artist One	artist1@example.com	pass1	Abstract painter	NULL	2025-12-05
2	Artist Two	artist2@example.com	pass2	Contemporary visual artist	NULL	2025-12-05
3	Artist Three	artist3@example.com	pass3	Landscape painter	NULL	2025-12-06
4	Artist Four	artist4@example.com	pass4	Modern art specialist	NULL	2025-12-06
5	Artist Five	artist5@example.com	pass5	Digital illustration artist	NULL	2025-12-06
6	Artist Six	artist6@example.com	pass6	Minimalist concept artist	NULL	2025-12-06
7	Artist Seven	artist7@example.com	pass7	Portrait and figure painter	NULL	2025-12-06
8	Artist Eight	artist8@example.com	pass8	Mixed media artist	NULL	2025-12-06
9	Artist Nine	artist9@example.com	pass9	Urban street style artist	NULL	2025-12-06
10	Artist Ten	artist10@example.com	pass10	Experimental abstraction	NULL	2025-12-06
11	Artist Eleven	artist11@example.com	pass11	Nature inspired compositions	NULL	2025-12-06
12	Artist Twelve	artist12@example.com	pass12	Contemporary conceptual works	NULL	2025-12-06

12 rows in set (0.00 sec)

3. Customer Table

```
mysql> SELECT * FROM Customer;
```

Customer_ID	Name	Email	Password	Address	Phone	Join_Date
1	Customer One	cust1@example.com	cpass1	Dhaka, Bangladesh	017XXXXXXX	2025-12-05
2	Customer Two	cust2@example.com	cpass2	Chittagong, Bangladesh	018XXXXXXX	2025-12-05
3	Customer Three	cust3@example.com	cpass3	Sylhet, Bangladesh	019XXXXXXX	2025-12-06
4	Customer Four	cust4@example.com	cpass4	Rajshahi, Bangladesh	016XXXXXXX	2025-12-06
5	Customer Five	cust5@example.com	cpass5	Khulna, Bangladesh	015XXXXXXX	2025-12-06
6	Customer Six	cust6@example.com	cpass6	Barishal, Bangladesh	017YYYYYYY	2025-12-06
7	Customer Seven	cust7@example.com	cpass7	Rangpur, Bangladesh	018YYYYYYY	2025-12-06
8	Customer Eight	cust8@example.com	cpass8	Mymensingh, Bangladesh	019YYYYYYY	2025-12-06
9	Customer Nine	cust9@example.com	cpass9	Comilla, Bangladesh	014XXXXXXX	2025-12-06
10	Customer Ten	cust10@example.com	cpass10	Bogura, Bangladesh	013XXXXXXX	2025-12-06
11	Customer Eleven	cust11@example.com	cpass11	Jessore, Bangladesh	017ZZZZZZZ	2025-12-06
12	Customer Twelve	cust12@example.com	cpass12	Noakhali, Bangladesh	018ZZZZZZZ	2025-12-06

12 rows in set (0.00 sec)

4. Category Table

```
mysql> SELECT * FROM Category;
```

Category_ID	Category_Name
1	Abstract
2	Portrait
3	Landscape
4	Modern
5	Digital Art
6	Street Art
7	Surrealism
8	Pop Art
9	Cubism
10	Realism
11	Black and White
12	Watercolor
13	Sculpture
14	Photography
15	Calligraphy

15 rows in set (0.00 sec)

5. Artwork Table

```
mysql> SELECT * FROM Artwork;
```

Artwork_ID	Artist_ID	Category_ID	Title	Description	Year	Price	Image_Path	Status	Upload_Date
1	1	1	Sunset Dreams	An abstract sunset in warm tones	2023	15000.00	sunset_dreams.jpg	Available	2025-12-05
2	1	2	Silent Portrait	Minimal portrait artwork	2022	12000.00	silent_portrait.jpg	Available	2025-12-05
3	2	3	Green Valley	Landscape with mountains and rivers	2021	18000.00	green_valley.jpg	Available	2025-12-05
4	1	4	City Lights	Modern city skyline at night	2023	20000.00	city_lights.jpg	Available	2025-12-06
5	2	5	Ocean Whisper	Calm seascape with soft waves	2022	17000.00	ocean_whisper.jpg	Available	2025-12-06
6	3	6	Silent Streets	Street art style composition	2023	19000.00	silent_streets.jpg	Available	2025-12-06
7	4	7	Dream Fragments	Surreal collage of dream elements	2021	22000.00	dream_fragments.jpg	Available	2025-12-06
8	5	8	Neon Pop	Pop art inspired neon colors	2024	25000.00	neon_pop.jpg	Available	2025-12-06
9	3	9	Broken Shapes	Cubist geometric figure study	2020	16000.00	broken_shapes.jpg	Available	2025-12-06
10	4	10	Quiet Morning	Realistic morning light in a room	2023	14000.00	quiet_morning.jpg	Available	2025-12-06
11	2	1	Crimson Horizon	Abstract red-toned horizon line	2024	15500.00	crimson_horizon.jpg	Available	2025-12-06
12	1	2	Faded Memories	Soft portrait with muted colors	2022	13000.00	faded_memories.jpg	Available	2025-12-06
13	5	3	Forest Echo	Landscape forest with misty hills	2021	18500.00	forest_echo.jpg	Available	2025-12-06

13 rows in set (0.00 sec)

6. Orders Table

```
mysql> SELECT * FROM Orders;
```

Order_ID	Customer_ID	Order_Date	Total_Amount	Status
1	1	2025-12-05 18:34:33	15000.00	Completed
2	2	2025-12-05 18:34:33	30000.00	Completed
3	1	2025-12-06 10:05:00	15000.00	Completed
4	2	2025-12-06 11:15:00	32000.00	Completed
5	3	2025-12-06 12:25:00	18500.00	Completed
6	4	2025-12-06 13:35:00	25000.00	Pending
7	5	2025-12-06 14:45:00	14000.00	Completed
8	6	2025-12-06 15:55:00	19000.00	Pending
9	2	2025-12-06 16:05:00	30000.00	Completed
10	3	2025-12-06 17:15:00	15500.00	Completed
11	4	2025-12-06 18:25:00	20000.00	Completed
12	5	2025-12-06 19:35:00	13000.00	Pending

12 rows in set (0.00 sec)

7. Order_Items Table

```
mysql> SELECT * FROM Order_Items;
```

Order_Item_ID	Order_ID	Artwork_ID	Price	Quantity
1	1	1	15000.00	1
2	2	2	12000.00	1
3	2	3	18000.00	1
4	3	4	15000.00	1
5	4	5	17000.00	1
6	4	6	15000.00	1
7	5	7	18500.00	1
8	6	8	25000.00	1
9	7	9	16000.00	1
10	8	10	14000.00	1
11	9	11	15500.00	1
12	10	12	13000.00	1
13	11	13	18500.00	1

13 rows in set (0.00 sec)

8. Exhibition Table

```
mysql> SELECT * FROM Exhibition;
```

Exhibition_ID	Title	Description	Start_Date	End_Date
1	Winter Art Fest	Group exhibition of winter-inspired artworks	2025-01-10	2025-01-30
2	Modern Expressions	Modern and abstract art showcase	2025-02-01	2025-02-20
3	Colors of Dhaka	Exhibition inspired by Dhaka city life	2025-03-01	2025-03-20
4	Shades of Silence	Minimal and monochrome works	2025-04-05	2025-04-25
5	Digital Horizons	Showcase of modern digital art	2025-05-01	2025-05-21
6	Nature Reimagined	Landscapes and nature-themed artworks	2025-06-10	2025-06-30
7	Faces and Stories	Portrait exhibition exploring human emotions	2025-07-05	2025-07-25
8	Abstract Realities	Non-figurative and experimental art	2025-08-01	2025-08-21
9	Urban Rhythm	Street art and city-inspired visuals	2025-09-10	2025-09-30
10	Water and Light	Watercolor and reflective surface studies	2025-10-05	2025-10-25
11	Sculpted Dimensions	Sculpture and 3D forms exhibition	2025-11-01	2025-11-21
12	Frames of Time	Photography exploring time and memory	2025-12-01	2025-12-20

12 rows in set (0.00 sec)

9. Exhibition_Artwork Table

```
mysql> SELECT * FROM Exhibition_Artwork;
```

Exhibition_ID	Artwork_ID
1	1
2	2
1	3
3	4
3	5
4	6
4	7
5	8
6	9
7	10
8	11
9	12
10	13

13 rows in set (0.04 sec)

2.7 Pattern Identification Queries

1. List all artworks with their artist name and category name.

```
SELECT
  aw.Artwork_ID,
  aw.Title,
  a.Name      AS Artist_Name,
  c.Category_Name
FROM Artwork aw
JOIN Artist a  ON aw.Artist_ID = a.Artist_ID
LEFT JOIN Category c ON aw.Category_ID = c.Category_ID
ORDER BY aw.Artwork_ID;
```

- 2. Find total money spent by each customer (only completed orders).**

```
SELECT
    c.Customer_ID,
    c.Name AS Customer_Name,
    SUM(o.Total_Amount) AS Total_Spent
FROM Customer c
JOIN Orders o ON c.Customer_ID = o.Customer_ID
WHERE o.Status = 'Completed'
GROUP BY c.Customer_ID, c.Name
ORDER BY Total_Spent DESC;
```

- 3. Show customers who placed orders worth more than 25,000 in total.**

```
SELECT
    c.Customer_ID,
    c.Name,
    SUM(o.Total_Amount) AS Total_Spent
FROM Customer c
JOIN Orders o ON c.Customer_ID = o.Customer_ID
WHERE o.Status = 'Completed'
GROUP BY c.Customer_ID, c.Name
HAVING SUM(o.Total_Amount) > 25000;
```

- 4. For each category, show average price of artworks and only keep categories with average price > 16,000.**

```
SELECT
    c.Category_Name,
    AVG(aw.Price) AS Avg_Price
FROM Category c
JOIN Artwork aw ON c.Category_ID = aw.Category_ID
GROUP BY c.Category_Name
HAVING AVG(aw.Price) > 16000
ORDER BY Avg_Price DESC;
```

- 5. List each exhibition with the number of artworks displayed in it.**

```
SELECT
    e.Exhibition_ID,
    e.Title,
    COUNT(ea.Artwork_ID) AS Total_Artworks
FROM Exhibition e
LEFT JOIN Exhibition_Artwork ea
```

```

    ON e.Exhibition_ID = ea.Exhibition_ID
GROUP BY e.Exhibition_ID, e.Title
ORDER BY Total_Artworks DESC;

```

6. Find each artist's total number of artworks that have been sold (appear in Order_Items).

```

SELECT
    a.Artist_ID,
    a.Name AS Artist_Name,
    COUNT(DISTINCT oi.Artwork_ID) AS Artworks_Sold
FROM Artist a
LEFT JOIN Artwork aw ON a.Artist_ID = aw.Artist_ID
LEFT JOIN Order_Items oi ON aw.Artwork_ID = oi.Artwork_ID
GROUP BY a.Artist_ID, a.Name
ORDER BY Artworks_Sold DESC;

```

7. Top 3 best-selling artworks by quantity sold.

```

SELECT
    aw.Artwork_ID,
    aw.Title,
    SUM(oi.Quantity) AS Total_Quantity_Sold
FROM Order_Items oi
JOIN Artwork aw ON oi.Artwork_ID = aw.Artwork_ID
GROUP BY aw.Artwork_ID, aw.Title
ORDER BY Total_Quantity_Sold DESC
LIMIT 3;

```

8. List artworks that have never been sold in any order.

```

SELECT
    aw.Artwork_ID,
    aw.Title,
    aw.Price
FROM Artwork aw
WHERE aw.Artwork_ID NOT IN (
    SELECT DISTINCT Artwork_ID FROM Order_Items
);

```

9. Find customers who have never placed any order.

```
SELECT
    c.Customer_ID,
    c.Name,
    c.Email
FROM Customer c
LEFT JOIN Orders o ON c.Customer_ID = o.Customer_ID
WHERE o.Order_ID IS NULL;
```

10. List exhibitions that showcase artworks from more than one artist.

```
SELECT
    e.Exhibition_ID,
    e.Title,
    COUNT(DISTINCT aw.Artist_ID) AS Distinct_Artists
FROM Exhibition e
JOIN Exhibition_Artwork ea ON e.Exhibition_ID = ea.Exhibition_ID
JOIN Artwork aw ON ea.Artwork_ID = aw.Artwork_ID
GROUP BY e.Exhibition_ID, e.Title
HAVING COUNT(DISTINCT aw.Artist_ID) > 1;
```

11. Show orders that contain two or more different artworks.

```
SELECT
    o.Order_ID,
    o.Customer_ID,
    COUNT(DISTINCT oi.Artwork_ID) AS Artwork_Count,
    o.Total_Amount
FROM Orders o
JOIN Order_Items oi ON o.Order_ID = oi.Order_ID
GROUP BY o.Order_ID, o.Customer_ID, o.Total_Amount
HAVING COUNT(DISTINCT oi.Artwork_ID) >= 2;
```

12. Monthly revenue summary (based on Order_Date).

```
SELECT
    DATE_FORMAT(o.Order_Date, '%Y-%m') AS YearMonth,
    SUM(o.Total_Amount) AS Total_Revenue,
    COUNT(*) AS Total_Orders
FROM Orders o
WHERE o.Status = 'Completed'
GROUP BY DATE_FORMAT(o.Order_Date, '%Y-%m')
ORDER BY YearMonth;
```

13. For each customer, show their latest order (date and amount).

```
SELECT
    c.Customer_ID,
    c.Name,
    o.Order_ID,
    o.Order_Date,
    o.Total_Amount
FROM Customer c
JOIN Orders o
    ON o.Customer_ID = c.Customer_ID
WHERE o.Order_Date = (
    SELECT MAX(o2.Order_Date)
    FROM Orders o2
    WHERE o2.Customer_ID = c.Customer_ID
);
```

14. Show total revenue generated by each artist from sold artworks.

```
SELECT
    a.Artist_ID,
    a.Name AS Artist_Name,
    SUM(oi.Price * oi.Quantity) AS Revenue_From_Sales
FROM Artist a
JOIN Artwork aw  ON a.Artist_ID = aw.Artist_ID
JOIN Order_Items oi ON aw.Artwork_ID = oi.Artwork_ID
JOIN Orders o    ON oi.Order_ID = o.Order_ID
WHERE o.Status = 'Completed'
GROUP BY a.Artist_ID, a.Name
ORDER BY Revenue_From_Sales DESC;
```

15. For each category, count how many artworks have appeared in at least one exhibition.

```
SELECT
    c.Category_Name,
    COUNT(DISTINCT aw.Artwork_ID) AS Artworks_In_Exhibitions
FROM Category c
JOIN Artwork aw      ON c.Category_ID = aw.Category_ID
JOIN Exhibition_Artwork ea ON aw.Artwork_ID = ea.Artwork_ID
GROUP BY c.Category_Name
ORDER BY Artworks_In_Exhibitions DESC;
```

PHASE 03 — PROJECT PROGRESS & CONTRIBUTION

3.1 Project Timeline Overview

The development of the **Art Gallery Management System (AGMS)** was carried out over multiple weeks during the Database Management System Laboratory course. The work was divided into clearly defined phases:

1. **Requirement Analysis & Planning**
2. **Database Design (ERD, Schema, Normalization)**
3. **Backend Implementation (MySQL + Python)**
4. **Frontend Development (HTML, CSS, JS)**
5. **Integration & Testing**
6. **Report Writing & Final Refinement**
7. **Making Presentation**

3.2 Group Progress Summary

The following table summarizes the overall contributions of the team members:

Sl. No.	Name	ID	Estimated Hours	Actual Hours	Main Responsibilities
1	Mutasim Billah	23549009057	40	46	Fronted Design, Backend Logic, Testing, DB Management
2	Abdullah Al Masum	23549009019	40	44	ERD, Report Making, Documentation, Presentation

3.3 Week-wise Progress (Group Level)

Week 1 – Topic Selection & Requirement Analysis

- Selected “Art Gallery Management System” as project topic
- Identified key users: Admin, Artist, Customer

- Listed core features and requirements
- Discussed initial system scope and limitations

Week 2 – Conceptual Design

- Identified entities and attributes
- Designed initial ER Diagram
- Defined relationships between Artist, Artwork, Customer, Orders, Exhibition
- Presented draft model for feedback

Week 3 – Logical Design & Normalization

- Converted ERD into relational schema
- Defined primary keys and foreign keys
- Performed normalization up to 3NF
- Prepared integrity constraints (domain, entity, referential)

Week 4 – Database Implementation & Sample Data

- Implemented tables in MySQL
- Applied constraints (PK, FK, UNIQUE, NOT NULL, CHECK)
- Inserted sample rows for testing
- Executed basic SELECT, JOIN, GROUP BY queries

Week 5 – Application Layer & UI

- Developed basic Flask routes (backend)
- Built HTML/CSS templates for main pages
- Integrated pages with database queries
- Implemented search, browse, login, upload artwork, and cart logic

Week 6 – Testing, Debugging & Reporting

- Tested all major functionalities: registration, login, CRUD operations, orders
- Fixed authentication, validation, and query bugs
- Finalized documentation and report structure
- Prepared the final presentation materials

3.4 Individual Progress Report – Abdullah Al Masum (ID: 23549009019)

3.4.1 Responsibilities

- Idea generation
- Writing SQL scripts (CREATE TABLE, INSERT, pattern queries)
- System testing and debugging
- Writing the report
- Making presentation

3.4.2 Work Summary Table

Time Period	Tasks Performed
Week 1	Helped analyze requirements, listed entities and attributes
Week 2	Designed ER Diagram draft, refined entities and relationships
Week 3	Performed schema refinement and normalization to 3NF
Week 4	Wrote MySQL CREATE TABLE scripts with constraints; populated sample data
Week 5	Checked routes for login, registration, artwork upload, cart, checkout
Week 6	Performed testing, fixed SQL and backend bugs, wrote database design & SQL sections of report

3.4.3 Skills & Learning

- Hands-on experience in ERD, relational schema design, normalization
- Practical understanding of connecting Python backend with MySQL
- Improved debugging skills for SQL and backend logic

3.5 Individual Progress Report – Mutasim Billah (ID: 23549009057)

3.5.1 Responsibilities

- Frontend UI design (HTML, CSS, layout)
- Backend design (Python, MySQL Connector)
- Integration of templates with backend routes

- Writing SELECT, JOIN, and pattern analysis queries
- Designing use case descriptions and screen descriptions

3.5.2 Work Summary Table

Time Period	Tasks Performed
Week 1	Contributed to requirement analysis and feature listing
Week 2	Helped finalize ERD; documented entity descriptions and relationships
Week 3	Assisted in defining relational schema and normalization examples
Week 4	Designed UI for Home, Login, Register, Browse Artwork, Artwork Details, Cart
Week 5	Integrated frontend templates with Flask routes, tested page navigation and forms
Week 6	Wrote functional requirement descriptions, screen overviews, and formatted the report for printing

3.5.3 Skills & Learning

- Improved UI/UX design skills using HTML/CSS
- Gained experience in integrating frontend with backend endpoints
- Learned how to write meaningful SQL queries for analytics

3.6 Challenges Faced

- Handling relational integrity while deleting records (e.g., deleting an artwork that is part of an order or exhibition)
- Debugging SQL syntax and foreign key constraint violations
- Designing a responsive interface that aligns well with database-driven content
- Managing time between lab tasks, project coding, and report writing

3.7 Mitigation Strategies

- Used incremental testing approach (test each module after implementation)
- Added careful foreign key definitions with appropriate cascading where safe
- Conducted cross-checks between ERD and implemented schema
- Divided responsibilities clearly between team members to avoid overlap

3.8 Overall Team Reflection

The team collaborated effectively and frequently discussed design decisions before implementation. Responsibilities were divided logically between database/backend and frontend/documentation. Regular testing after each phase reduced the number of critical issues toward the end. The project provided a realistic experience of how DBMS concepts translate into a fully functional application.

PHASE 04 — FINAL SECTIONS

4.1 Limitations of the System

Although the Art Gallery Management System (AGMS) successfully implements core gallery management functions, several constraints limit the system's full capabilities in its current prototype form:

Technical Limitations

1. Local Image Storage

Artwork images are stored locally; no cloud integration is implemented for scalable storage.

2. No Real Payment Gateway

Purchase transactions are simulated. Integration with real-world payment services (e.g., SSLCommerz, Stripe, PayPal) is not included.

3. Limited Recommendation System

The system does not currently analyze user behavior to recommend artworks or exhibitions.

4. No Multi-Admin Privilege Levels

All admins share the same privilege level instead of having hierarchical access control.

5. **No Automated Notification System**

Users (artists/customers) do not receive automated email/SMS updates about orders or exhibition invitations.

User Experience Limitations

1. **Basic UI/UX**

Although functional, the interface could be more modern and optimized for mobile responsiveness.

2. **Lack of Advanced Search Filters**

Search options are limited; filters such as color theme, era, or multi-criteria sorting are not available.

3. **Exhibition Virtual Tour Missing**

The system does not include a virtual display or AR/VR integration.

Database Limitations

1. **No Backup/Restore Automation**

Database backup must be done manually.

2. **Limited Transaction Support**

The current system does not handle complex rollback-recovery transactions.

3. **Analytics Are Limited**

While basic reporting is available, advanced analytics such as artist ranking trends or customer buying patterns are not implemented.

4.2 Future Improvements

AGMS can be expanded and enhanced significantly. Below are potential upgrades for future versions:

Technical Enhancements

- **Cloud Storage Integration** for artwork images
- **Real Payment Gateway Integration**
- **AI-Based Artwork Recommendation System**
- **Role-Based Admin Privileges** (Super Admin, Moderator, Exhibition Manager)

User-Oriented Enhancements

- **Mobile App Version** (Android & iOS)
- **Dark Mode & Modern UI Enhancements**
- **Live Chat Between Artists and Customers**
- **Wishlist / Favorites Feature**
- **Review & Ratings System**

Gallery & Exhibition Enhancements

- **Virtual Gallery Walkthrough (3D / VR)**
- **Automated Exhibition Scheduling**
- **Artist Collaboration Modules**

Database-Level Enhancements

- **Implement Stored Procedures, Triggers, Views, and Transaction Handling**
- **Automatic database backup & restore** utilities
- **Advanced BI Dashboards** for analytics

4.3 Learning Outcomes

Working on the Art Gallery Management System allowed the team to develop strong practical knowledge in several key areas of database management, application development, and teamwork.

Database Concepts Learned

- Designing an ER Diagram based on real-world requirements
- Converting ER models into normalized relational schemas
- Ensuring data integrity using primary keys, foreign keys, and check constraints
- Writing efficient SQL queries including JOINS, GROUP BY, aggregated queries, and pattern identification queries
- Understanding normalization from 1NF to 3NF

Backend & Application Development Skills

- Connecting Python Flask with a MySQL database
- Implementing CRUD operations using SQL queries in backend routes
- Handling session management, authentication, and user roles
- Validating user inputs and preventing SQL injection

Frontend Development Skills

- Building structured UI layouts using HTML and CSS
- Creating dynamic forms, detail pages, and catalog pages
- Ensuring responsive and user-friendly design

Soft Skills & Team Experience

- Effective communication and task distribution
- Time management across multiple development phases
- Writing formal documentation according to academic standards
- Collaborative debugging and problem-solving

4.4 Conclusion

The Art Gallery Management System (AGMS) provides an effective and structured solution for managing artworks, artists, customers, and exhibition activities. By integrating a relational database with a responsive web interface, AGMS demonstrates how database management systems can support real-world applications through data integrity, consistency, and automation.

Through the phases of requirement analysis, database design, normalization, implementation, testing, and documentation, the project strengthened our understanding of database principles and their role in application development. Although the system has limitations, it establishes a strong foundation for further enhancements such as AI-based recommendations, real payment systems, and cloud integration.

Overall, AGMS successfully meets the intended objectives of the course project by providing a functional, well-designed, and academically structured database-backed application.